

SQL Joins

Name: Nomit Bagaria

Assignment Name: SQL Joins Assignment

Mob.No: 9828255290

Email: bagarianomit@gmail.com

Creating given dummy tables in MySQL Workbench using given data :

Table 1: Customers

```
CREATE TABLE Customers (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(50),  
    City VARCHAR(50)  
);
```

```
INSERT INTO Customers  
VALUES  
(1,'John Smith','New York'),  
(2,'Mary Johnson','Chicago'),  
(3,'Peter Adams','Los Angeles'),  
(4,'Nancy Miller','Houston'),  
(5,'Robert White','Miami');
```

Table 2: Orders

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    CustomerID INT,  
    OrderDate DATE,  
    Amount INT  
);
```

```
INSERT INTO Orders  
VALUES  
(101,1,'2024-10-01',250),  
(102,2,'2024-10-05',300),
```

```
(103,1,'2024-10-07',150),
(104,3,'2024-10-10',450),
(105,6,'2024-10-12',400);
```

Table 3: Payments

```
CREATE TABLE Payments (
  PaymentID VARCHAR(10) PRIMARY KEY,
  CustomerID INT,
  PaymentDate DATE,
  Amount INT
);
```

```
INSERT INTO Payments
VALUES
('P001',1,'2024-10-02',250),
('P002',2,'2024-10-06',300),
('P003',3,'2024-10-11',450),
('P004',4,'2024-10-15',200);
```

Table 4: Employees

```
CREATE TABLE Employees (
  EmployeeID INT PRIMARY KEY,
  EmployeeName VARCHAR(50),
  ManagerID INT
);
```

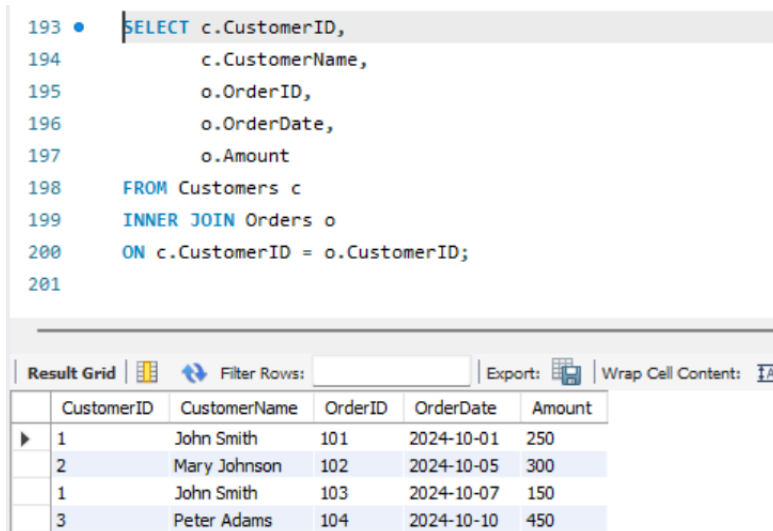
```
INSERT INTO Employees
VALUES
(1,'Alex Green',NULL),
(2,'Brian Lee',1),
(3,'Carol Ray',1),
(4,'David Kim',2),
(5,'Eva Smith',2);
```

✓	46	00:54:50	CREATE TABLE Customers (CustomerID INT PRIMARY KEY, CustomerNa...	0 row(s) affected	0.016 sec
✓	47	00:55:03	INSERT INTO Customers VALUES (1,'John Smith','New York'), (2,'Mary Johnson'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
✓	48	00:55:09	CREATE TABLE Orders (OrderID INT PRIMARY KEY, CustomerID INT, ...	0 row(s) affected	0.094 sec
✓	49	00:55:18	INSERT INTO Orders VALUES (101,1,'2024-10-01',250), (102,2,'2024-10-05',300...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
✓	50	00:55:25	CREATE TABLE Payments (PaymentID VARCHAR(10) PRIMARY KEY, Cu...	0 row(s) affected	0.016 sec
✓	51	00:55:33	INSERT INTO Payments VALUES ('P001',1,'2024-10-02',250), ('P002',2,'2024-10...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.000 sec
✓	52	00:55:39	CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, EmployeeN...	0 row(s) affected	0.016 sec
✓	53	00:55:48	INSERT INTO Employees VALUES (1,'Alex Green',NULL), (2,'Brian Lee',1), (3,'C...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.062 sec

Q1. Retrieve all customers who have placed at least one order.

Answer:

```
SELECT c.CustomerID,  
       c.CustomerName,  
       o.OrderID,  
       o.OrderDate,  
       o.Amount  
FROM Customers c  
INNER JOIN Orders o  
ON c.CustomerID = o.CustomerID;
```

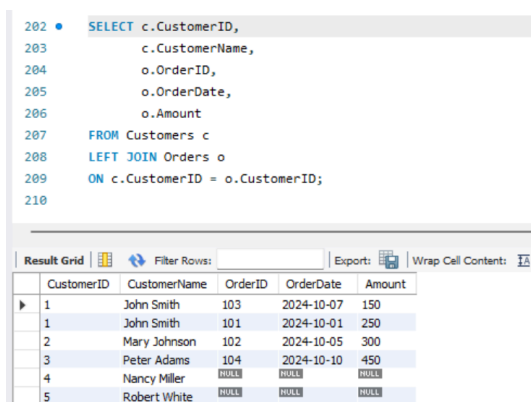


CustomerID	CustomerName	OrderID	OrderDate	Amount
1	John Smith	101	2024-10-01	250
2	Mary Johnson	102	2024-10-05	300
1	John Smith	103	2024-10-07	150
3	Peter Adams	104	2024-10-10	450

Q2. Retrieve all customers and their orders, including customers who have not placed any orders.

Answer:

```
SELECT c.CustomerID,  
       c.CustomerName,  
       o.OrderID,  
       o.OrderDate,  
       o.Amount  
FROM Customers c  
LEFT JOIN Orders o  
ON c.CustomerID = o.CustomerID;
```

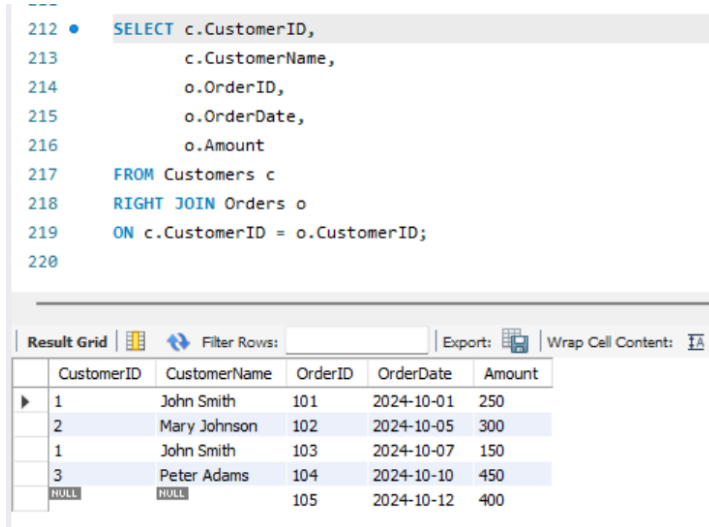


CustomerID	CustomerName	OrderID	OrderDate	Amount
1	John Smith	103	2024-10-07	150
1	John Smith	101	2024-10-01	250
2	Mary Johnson	102	2024-10-05	300
3	Peter Adams	104	2024-10-10	450
4	Nancy Miller	NULL	NULL	NULL
5	Robert White	NULL	NULL	NULL

Q3.Retrieve all orders and their corresponding customers, including orders placed by unknown customers.

Answer:

```
SELECT c.CustomerID,  
       c.CustomerName,  
       o.OrderID,  
       o.OrderDate,  
       o.Amount  
FROM Customers c  
RIGHT JOIN Orders o  
ON c.CustomerID = o.CustomerID;
```



The screenshot shows a SQL query editor with a query window and a results grid. The query is a RIGHT JOIN between Customers and Orders. The results grid displays the following data:

	CustomerID	CustomerName	OrderID	OrderDate	Amount
1	1	John Smith	101	2024-10-01	250
2	2	Mary Johnson	102	2024-10-05	300
	1	John Smith	103	2024-10-07	150
3	3	Peter Adams	104	2024-10-10	450
	NULL	NULL	105	2024-10-12	400

Q4. Display all customers and orders, whether matched or not.

Answer:

```
SELECT c.CustomerID,  
       c.CustomerName,  
       o.OrderID,  
       o.OrderDate,  
       o.Amount  
FROM Customers c  
LEFT JOIN Orders o  
ON c.CustomerID = o.CustomerID
```

UNION

```
SELECT c.CustomerID,  
       c.CustomerName,  
       o.OrderID,
```

```

o.OrderDate,
o.Amount
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;

```

231
232 SELECT c.CustomerID,
233 c.CustomerName,
234 o.OrderID,
235 o.OrderDate

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	CustomerID	CustomerName	OrderID	OrderDate	Amount
▶	1	John Smith	103	2024-10-07	150
	1	John Smith	101	2024-10-01	250
	2	Mary Johnson	102	2024-10-05	300
	3	Peter Adams	104	2024-10-10	450
	4	Nancy Miller	NULL	NULL	NULL
	5	Robert White	NULL	NULL	NULL
	NULL	NULL	105	2024-10-12	400

Q5. Find customers who have not placed any orders.

Answer:

```

SELECT c.CustomerID,
       c.CustomerName
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;

```

241 • SELECT c.CustomerID,
242 c.CustomerName
243 FROM Customers c
244 LEFT JOIN Orders o
245 ON c.CustomerID = o.CustomerID
246 WHERE o.OrderID IS NULL;
247

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	CustomerID	CustomerName
▶	4	Nancy Miller
	5	Robert White

Q6. Retrieve customers who made payments but did not place any orders.

Answer:

```

SELECT c.CustomerID,
       c.CustomerName
FROM Customers c
INNER JOIN Payments p

```

```

ON c.CustomerID = p.CustomerID
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;

```

```

248 • SELECT c.CustomerID,
249         c.CustomerName
250 FROM Customers c
251 INNER JOIN Payments p
252 ON c.CustomerID = p.CustomerID
253 LEFT JOIN Orders o
254 ON c.CustomerID = o.CustomerID
255 WHERE o.OrderID IS NULL;
256

```

CustomerID	CustomerName
4	Nancy Miller

Q7. Generate a list of all possible combinations between Customers and Orders

Answer:

```

SELECT c.CustomerName,
       o.OrderID
FROM Customers c
CROSS JOIN Orders o;

```

CustomerName	OrderID
Robert White	101
Nancy Miller	101
Peter Adams	101
Mary Johnson	101
John Smith	101
Robert White	102
Nancy Miller	102
Peter Adams	102
Mary Johnson	102
John Smith	102

```

257 • SELECT c.CustomerName,
258         o.OrderID
259 FROM Customers c
260 CROSS JOIN Orders o;
261

```

CustomerName	OrderID
Robert White	101
Nancy Miller	101
Peter Adams	101
Mary Johnson	101
John Smith	101
Robert White	102
Nancy Miller	102
Peter Adams	102
Mary Johnson	102
John Smith	102
Robert White	103
Nancy Miller	103
Peter Adams	103
Mary Johnson	103
John Smith	103
Robert White	104
Nancy Miller	104
Peter Adams	104
Mary Johnson	104

Robert White	103
Nancy Miller	103
Peter Adams	103
Mary Johnson	103
John Smith	103
Robert White	104
Nancy Miller	104
Peter Adams	104
Mary Johnson	104
John Smith	104
Robert White	105
Nancy Miller	105
Peter Adams	105
Mary Johnson	105
John Smith	105

Q8. Show all customers along with order and payment amounts in one table.

Answer:

```

SELECT c.CustomerID,
       c.CustomerName,
       o.OrderID,
       o.Amount AS OrderAmount,
       p.PaymentID,
       p.Amount AS PaymentAmount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
LEFT JOIN Payments p
ON c.CustomerID = p.CustomerID;

```

```

262 • SELECT c.CustomerID,
263         c.CustomerName,
264         o.OrderID,
265         o.Amount AS OrderAmount,
266         p.PaymentID,
267         p.Amount AS PaymentAmount
268 FROM Customers c
269 LEFT JOIN Orders o
270 ON c.CustomerID = o.CustomerID
271 LEFT JOIN Payments p
272 ON c.CustomerID = p.CustomerID;
273

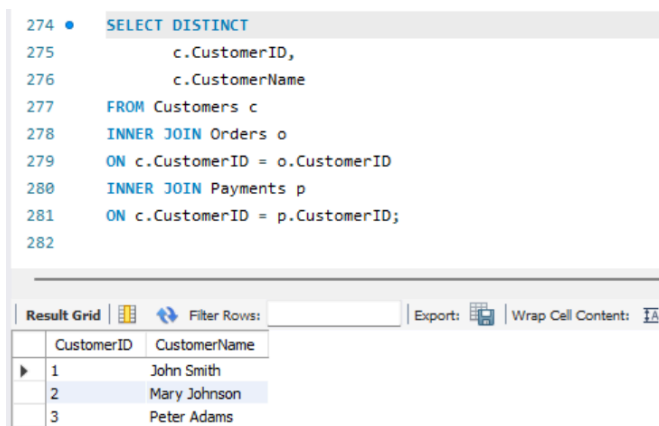
```

CustomerID	CustomerName	OrderID	OrderAmount	PaymentID	PaymentAmount
1	John Smith	103	150	P001	250
1	John Smith	101	250	P001	250
2	Mary Johnson	102	300	P002	300
3	Peter Adams	104	450	P003	450
4	Nancy Miller	NULL	NULL	P004	200
5	Robert White	NULL	NULL	NULL	NULL

Q9. Retrieve all customers who have both placed orders and made payments.

Answer:

```
SELECT DISTINCT
    c.CustomerID,
    c.CustomerName
FROM Customers c
INNER JOIN Orders o
ON c.CustomerID = o.CustomerID
INNER JOIN Payments p
ON c.CustomerID = p.CustomerID;
```



The screenshot shows a SQL query editor with a query that has been executed. Below the query, there is a 'Result Grid' tab. The query is as follows:

```
274 • SELECT DISTINCT
275     c.CustomerID,
276     c.CustomerName
277 FROM Customers c
278 INNER JOIN Orders o
279 ON c.CustomerID = o.CustomerID
280 INNER JOIN Payments p
281 ON c.CustomerID = p.CustomerID;
282
```

The 'Result Grid' tab is active, showing a table with two columns: 'CustomerID' and 'CustomerName'. The table contains three rows of data:

	CustomerID	CustomerName
▶	1	John Smith
	2	Mary Johnson
	3	Peter Adams